

Oct.-16/T.E./Insem.-46

T.E. (Information Technology) (Semester - I)**THEORY OF COMPUTATION****(2012 Pattern)***Time : 1 Hour]**[Max. Marks : 30**Instructions to the candidates:*

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 of Q.6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data if necessary.*

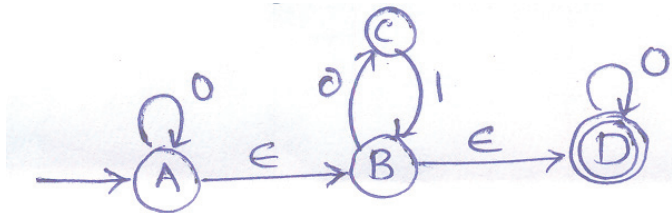
Unit - I**Q1) a) Give formal definitions for the following [5]**

- i) NFA with ϵ - transitions
- ii) ϵ -closure

b) Design an FA over $\Sigma = \{0,1\}$ for the following [5]

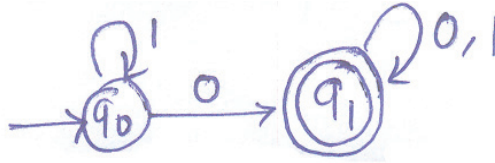
- i) Strings which end in either "00" or "11"
- ii) Strings which contain either "01" or "110"

OR

Q2) a) Compare Moore and Mealy machines with suitable example. [5]**b) Convert the following NFA- ϵ to equivalent NFA. [5]****P.T.O.**

Unit - II

- Q3) a)** Construct a Regular Expression corresponding to the following transition diagram using Arden's theorem. [5]



- b) Write a short note on the applications of regular expressions. [5]

OR

- Q4) a)** Construct a DFA that accepts the language represented by $0^*1^*2^*$. Make use of NFA. [8]
- b) Define pumping lemma for regular languages. [2]

Unit - III

- Q5) a)** Write CFG for the following languages [5]
- i) $L = \{a^n b^n \mid n \geq 1\}$
 - ii) $(baa + abb)^*$
- b) Simplify the given grammar [5]
- $S \rightarrow aC \mid SB$
 $A \rightarrow bSCa$
 $B \rightarrow aSB \mid bBC$
 $C \rightarrow aBC \mid ad$

OR

- Q6) a)** Define the following with suitable example [6]
- i) Regular grammar
 - ii) Ambiguous grammar
- b) Convert the following grammar to CNF [4]
- $S \rightarrow Aba \mid aab$
 $A \rightarrow aS$

